

All problems are from the end of chapter 1 in Artin's Algebra, 2nd edition **published by PHI**. *Be sure to use the correct edition as the problems are somewhat different in different editions. See the image of the cover attached at the end.*

HW is critical for your learning, so do follow the rules explained the email, which are reproduced below.

=====

Important points: (i) Discussion / any genuine collaboration is allowed (even encouraged), but copying is not (even copying from a conversation). A good procedure is to write down your final answers sitting alone using only your mind and without accessing any text, notes (even your own), conversation, internet resource or anything else. Submit only what you understand fully and can recreate on the spot if asked.

(ii) Quality of writing is of paramount importance. Write coherently using full sentences. Math notation is fine (including for verbs such as $=$ or $<$) but it still needs to be in a sentence. Justify appropriately where necessary, even in computational problems. You do not need to repeat the explanations done in the text/class: it is enough to just write a clear reference to the appropriate fact (but do revise in your mind the full underlying explanation until it flows smoothly in your mind).

(iii) While you are required to hand in only the assigned problems, the standing instruction and my expectation is that you should be able to solve ALL problems from relevant sections of Artin (with the exception of those that I label as optional). As an in-between task, I will generally list some suggested problems for extra practice. However, if required, you should solve more problems until you gain mastery.

=====

HW1 Write the problems in given order.

- **Four problems from 1.1**

6+10 (Describe each of AD and DA in a sentence. No need to justify.)

13

14

Find each 2×2 matrix that commutes with ALL 2×2 matrices. It may help to do at least parts of problem 12 first.

Suggested: 7,8,9,11,15

- **Three problems from 1.2**

2

6 (Use row operations.)

8

Suggested: 1,3,4,5,9,10 (Do enough of these computations to gain fluency.)

- **No assigned problem from 1.3**

All 4 problems are suggested. 1 and 2 are quick by appealing to 1.3.1. Last two (3 and 4) are open ended, so see what you can say. Optional variation for 4: what if only integers are allowed as matrix entries and scalars? Again row/column operations must be reversible.

- **One problem from Miscellaneous**

3+4 (Why is 4 clubbed with 3?)

Suggested: 2,8. Optional: 10,11 (Tell me after class if you do these.)

=====

HW2 Write the problems in given order.

- **Five problems from the end of Chapter 3**

2 in "Vector Spaces" + 1 in "Bases and dimension" (Take $F = \text{real numbers}$ if you like.)

"Bases and Dimension": 3 (What does it mean for a function to be 0?), 7

"Computing with Bases": 1ab (Find the easiest way for part a), 3 (Assemble vectors in B as columns of a matrix. What does each of the three given operations do to this matrix?)

- **Three more problems given below**

For 7 and 8 you may use only the following: everything written on the first page of Lecture 8 notes and the following three facts. $\text{RREF}(A) = EA$ where E is a product of elementary matrices. Each elementary matrix has an inverse which is also an elementary matrix. For invertible n by n matrices P and Q , the product PQ is invertible (the inverse being easy to write down).

Important note: (1) Whenever you use any of the above in your writing, you must specify precisely which statement you are using. In general whenever you are appealing to some standard result in your reasoning, you must name it or otherwise make it crystal clear what you are using. (2) While you need not justify anything in the above list in your written work, you **must** be able to completely justify ALL of it to yourself from first principles. How will you complete the 5th entry in each half of the table in the notes?

6) Let $A =$ the matrix with 3 rows and 5 columns having first row 1 2 3 4 5, second row 10 20 30 40 50 and third row 3 6 7 18 14.

Using only row operations on A , find a basis for (i) Null space of A (ii) Column space of A (iii) Row space of A . In each case give a succinct justification for your procedure: why vectors in your claimed basis are in the respective space, why they are linearly independent and why they span. In some parts some of these points may be obvious.

7) Let A be an $r \times c$ matrix with the associated function $f(x) = Ax$. Show that any two the following three statements implies the third.

- (i) A is a square matrix, i.e., $r = c$.
- (ii) $Ax = 0$ has only the trivial solution (i.e., f is injective).
- (iii) $Ax = b$ has a solution for every vector b (i.e., f is surjective).

If you use the table in Lecture 8 notes well, the solution will be short. The "i.e. ..." statements are part this table so you do not need to prove them. They are included only as important reminders and to emphasize analogy with a function between finite sets.

8) This is an extremely important problem!

Let A be a square matrix. Show that the following are equivalent. You may use the given outline, but before seeing it, first try the problem on your own and then compare. You may also give your own (CLEAN) proof.

(a) A is invertible. (b) The function $f(x) = Ax$ is a bijection. (c) There exists matrix L such that $LA = I$. (d) There exists matrix R such that $AR = I$. (e) $\text{RREF}(A) = \text{identity matrix } I$. (f) A is a product of elementary matrices.

Outline

If A is invertible, clearly (c) and (d) are true. Leave (b) for now.

If (c) is true, suppose $Ax = 0$. Then ... Therefore $\text{RREF}(A) = I$

If (d) is true, use parallel statement in the table to get $\text{RREF}(A) = I$.

$\text{RREF}(A) = I$ means we can write ... = I , where E is Therefore $A = \dots$. RHS is a product of elementary matrices because

Any product of elementary matrices is invertible because ...

This proves the equivalence of everything except (b).

Show that (a) easily implies (b). Which statement does (b) imply easily?

The notes below are not part of the required homework. They are included to encourage you to get the most out of this problem.

Note 1: Draw a diagram of implications among (a) to (f) proved above. If you place the letters right on the page, it should help in fixing the flow of logic in your mind, which is the point of this problem. You should think about it enough so that you are able to instantly recreate all statements and efficiently prove the equivalence of any two.

Note 2: Prove that one can add two more equivalent statements. (g) The function f is injective. (h) f is surjective. Where do these two statements fit into the diagram? Hint: using problem 7, see that any one of (c) and (d) also implies (b), though this is less direct than (a) implying (b).

Note 3: Conceptually (a) and (b) are closest and should be directly equivalent. That (b) implies (a) is not (yet) immediate for the following reason. Bijectivity of $f(x) = Ax$ does mean that the inverse function of f exists, but we don't know how to relate this inverse function to a matrix (which then will turn out to be the sought for inverse of A). The issue is more fundamental than invertibility: once we understand that matrices are "one and the same" as linear maps, it will give an immediate conceptual proof of equivalence of (a) and (b), because inverse of an isomorphism is an isomorphism (as we saw). For now, we took a scenic tour through row reduction and elementary matrices to get to (a) from (b).

Note 4: For a square matrix, all 10 statements in the Lecture 8 table are equivalent and can be added to the ones listed in this problem. Two of these are (g) and (h) in Note 2. The statements about there being a pivot in each row/column are directly equivalent (for a square matrix A) to $\text{RREF}(A) = I$. The remaining 6 statements are in terms of solutions of equations or vector space language (linear independence, span). You should also see that these last ones can be stated in terms basis/dimension. We will add a few more important equivalent statements in terms of the determinant and eigenvalues. Stay tuned!

Keep returning to this problem periodically to ensure that it (along with the notes) stays fresh in your mind. ***Did I mention that this is important?***

• Suggested

ALL remaining problems from "Vector spaces" and "Bases and dimension" sections. Most of these are easy/discussed in class. Problem 5 in "Bases and dimension" will likely be assigned at some point. We will get to the remaining problems in

"Computing with bases" later.

Highly recommended: Read Strang's section 3.5 "Dimensions of the four subspaces" for a friendly and thorough discussion AND solve several problems of your choice given at the end of the section. In problem 6 above we have ignored for now Strang's fourth subspace: (iv) null space of the transpose of A. To calculate its basis and dimension, of course you can apply the procedure in part (i) of problem 6 to the transpose of A. That will effectively mean doing column operations on A. Can you instead do it using row operations on A?

- **Optional**

Succinctly prove that any matrix A has a unique RREF by showing how to directly write $\text{RREF}(A)$ by using the solution set of $Ax = 0$. Here is a possible outline that you can fill with justification.

First identify pivot columns as follows. The pivot variables are those x_i for which the following is true: for any solution with $x_k = (\text{what})$ for all k (related how to i), x_i must satisfy (what). Once pivot columns are decided, the potentially nonzero entries in other columns are obtained as follows: the entry in pq slot is obtained from the unique value of (which variable) when (which variables are assigned which values).

=====

HW3 Write the problems in given order.

- **1 problem from the end of chapter 3**

Computing with Bases: 1c + 2 ab (Calculate directly from definition)

- **5 problems from the end of chapter 4**

The Dimension Formula: 4 (Earlier problem 5 was also assigned, but it was discussed in class, so I am removing it. To replace it I am adding one more problem at the end.)

Matrix of a Linear Transformation: 1 (Use the following notation for entries of A and B: first row of A = a_{11} a_{12} etc.), 4

Linear Operators: 1, 3

- **The following two problems**

7) Let V and W be arbitrary vector spaces.

(a) Show that $\text{Hom}(V, W)$ is a vector space under pointwise addition and scalar multiplication.

(b) Suppose $B = \{v_1, \dots, v_p\}$ is a basis for V and $C = \{w_1, \dots, w_q\}$ is a basis for W. For any linear map T from V to W, we constructed a q by p matrix with respect to given bases. Call this matrix $f(T)$. Show that f is an isomorphism from the vector space $\text{Hom}(V, W)$ to the vector space of all q by p matrices with real entries.

(c) Take T_1 and T_2 in $\text{Hom}(V, W)$ and S in $\text{Hom}(W, U)$ for some vector space U. Show that $(S \text{ composed with } (T_1 + T_2)) = (S \text{ composed with } T_1) + (S \text{ composed with } T_2)$. State without proof analogous formula using R in $\text{Hom}(U, V)$. In light of what we did in class (about composition of linear maps vis a vis their matrices) and part (b) above, these formulas translate into a property of matrix operations. State

this property and explain very briefly.

8) (a) Let V, W, U be finite dimensional vector spaces, T a linear map from V to W and S linear map from W to U . Thus ST (i.e. the composition T followed by S) is a linear map from V to U . Prove using only abstract vector space language that $\text{rank}(ST)$ is less than or equal to $\text{rank}(S)$ as well as $\text{rank}(T)$.

(b) Let A and B be matrices such that the product AB is defined. Prove using only matrices (without any mention of vector spaces) that $\text{rank}(AB)$ is less than or equal to $\text{rank}(A)$ as well as $\text{rank}(B)$. One possible line of thought: the vector Ax is a linear combination of the columns of A (with which scalars?). What can you say about column vectors of AB ? About row vectors of AB ?

(c) Your proofs in (a) and (b) should be independent of each other. Now explain briefly but precisely how either of (a) and (b) can be deduced from the other.

Suggested

All remaining problems for first three sections of chapter 4 (except problem 3.2: we will see similarity and conjugation later). Can you describe matrices in the answer to 2.3 geometrically as a subset of $\mathbb{R}^4$? Do try 2.5. We did 1.5 in class. All others can be done orally!

Highly recommended: chapter 4 M.5, chapter 3 M.2 (this will be very important for us later) and M.4.

In problem 8 above, can you find interesting condition(s) on A or S that imply equality? Likewise for B or T .

=====

HW4 Write the problems in given order.

- **Two problems from the end of chapter 3**

Direct Sums: 1+2

Infinite-Dimensional Spaces: 1 + also find a vector that is not in the span of the set $\{w, e_1, e_2, \dots\}$.

- **Five problems from the end of chapter 1**

Determinants: 3 (this is called the Cartan matrix of type A), 6. Also do 1d in your head instantaneously but do not submit it.

Other formulas for determinant: 1 only for the fourth matrix (the one with top row a, b, c), 2

Miscellaneous: 7

- **The following problem**

8) Let A = the 3×5 matrix from HW 2 having first row 1 2 3 4 5, second row 10 20 30 40 50 and third row 3 6 7 18 14. Consider the map $T(x) = Ax$ from $\mathbb{R}^5$ to $\mathbb{R}^3$. Find an explicit basis of $\mathbb{R}^5$ and an explicit basis of $\mathbb{R}^3$ such that with respect to these bases the matrix of T has matrix M = identity of suitable size in top left corner and zeros everywhere else.

Your final answer must be two lists of vectors with actual numbers in them. It should not be expressed in theoretical language such as "extend the basis in any way". If you wish, you may refer to your work in HW 2 where you found bases for various spaces associated to A . You need not repeat any work. You may just take the results from your earlier work.

Suggested

All remaining problems from these sections except the two starred problems from Infinite dimensional spaces, which are optional.

In problem 8 above find matrices Q and P such that $(Q \text{ inverse}) A P = M$.

=====

HW5 Write the problems in given order.

- **Five problems from the end of chapter 3**

Fields: 5,6 (You may refer to work from one subpart in another if relevant).

Computing with bases: 4, 5a (Note: The general linear group $GL_n(F)$ is the set of all invertible $n \times n$ matrices with entries in the field F . Special linear group is the set of matrices with determinant 1.)

Miscellaneous: M.1

- **Three problems from the end of chapter 4**

All three problems are purely computational, but do justify each computational step to yourself even if you do not write the reasoning.

Eigenvectors: 4

Characteristic polynomial: 1(a), 2. Regard t as an unknown constant while calculating the determinant of the relevant matrix. (Remark: So strictly speaking what you get is a polynomial function rather than a polynomial with symbol t . If this distinction does not make sense to you, you may ignore this entire remark without any loss.)

- **Do for yourself but do not turn in.**

Read all your work in HW 1 to 4 and see that all the theory and methods stay valid over arbitrary fields. Identify the few problems where some modification in language/conclusion is required.

Strongly suggested from Chapter 3: 1 to 4 from Fields, 4 from Computing with bases for GL_n and SL_n .

Suggested from Chapter 4: 2 from Linear operators. Try to identify easy problems from those on the sections Eigenvectors and Characteristic polynomial.

Optional: 7 to 11 from Fields, M.5. Suppose a field K has a subfield E , which has a subfield F . Show that $(\text{dimension of } K \text{ over } F) = (\text{dimension of } K \text{ over } E)(\text{dimension of } E \text{ over } F)$. All of this is part of later algebra courses, but feasible for you to do now. Also try 5b from Computing with bases and then the general case of k -dimensional subspaces in $(F_p)^n$.

HW 6: optional in the sense that best 5 of the 6 HW will count.

HW Deadline: Mon Nov 28, but if you have not solved many problems from Artin 4.4 to 4.7, then you should really do the required part of this HW for yourself before the quiz and certainly before the final exam. A good test of your preparation is whether you can solve with *ease* the problems recommended below in **0**.

Notation: Let $J(\lambda, k)$ denote the $k \times k$ Jordan block with eigenvalue λ , i.e., the matrix in which each of the k diagonal entries is λ , each of the $k - 1$ entries in slots $(i, i + 1)$ is 1 and all other entries are 0. As a shorthand notation, one can write a general matrix in Jordan form in a single line as direct sum of various $J(\lambda, k)$, i.e., $(J(\lambda_1, k_1) \oplus J(\lambda_1, k_2) \oplus \cdots) \oplus (\text{Jordan blocks for } \lambda_2) \oplus \cdots$. For uniqueness, arrange Jordan blocks for each eigenvalue in decreasing order according to size, i.e., $k_1 \geq k_2 \geq \cdots$ and similarly for other eigenvalues.

0. These problems are strongly recommended and should be quick work. Do NOT turn in any of these except the few that are listed below. (A) indicates problems that you should be able to do orally or by short straightforward computation. (B) indicates worthwhile problems that may require a little more work but are still fairly easy. (The remaining problems from these sections – primarily a handful in 4.5 – mostly just require more computation. You can do them later if you are interested.)

Section 4.4	(A) 1,3,4,8	(B) 2,5,6
Section 4.5	(A) 1,2,3,7	(B) 6
Section 4.6	(A) 1,2,3,5,6,7,8	(B) 4,9,10
Section 4.7	(A) 1,2,3,4,5,7	(B) 6,8,9
Section 5.1	(A) 2,4	(B) 1

Problems to write (if you choose to submit this HW). This homework should not require a lot of writing because all problems except problem 1 are very concrete/computational, but while writing the solutions you should still justify your answers in a few crisp sentences by appealing to appropriate theorems.

1. A projection is an operator T on a vector space V such that the following happens. There are two nonzero T -invariant subspaces K and R with $V = K \oplus R$ such that $T|_K = 0$ and $T|_R = id_R$. Fill in the blanks and prove the statements: If T is a projection then K is its _____ and R is its _____. T is a projection if and only if the minimal polynomial of T is _____. (Compare problem 7 in 4.7.)

2. Problem 6 in 4.5.

3. Problem 4 in 4.6.

4. For a vector space V with basis $\{u, v, w\}$, consider the linear operator T defined by $T(u) = v, T(v) = w$ and $T(w) = v$. Write the matrix A of T with respect to the given basis. Take the underlying field to be in turn $\mathbb{R}, \mathbb{C}, \mathbb{F}_p$, where p is a prime number. For each of these cases, find the minimal polynomial of T , decide if T is diagonalizable and if yes, explicitly diagonalize A .

5. Problem 6 in 4.7. Then look at problem 6 in 4.4. Is it the same?

6. Show that there is a unique similarity class of nilpotent matrices N with complex entries with the following properties: $\dim(\ker N) = 4, \dim(\ker N^2) = 8, \dim(\ker N^3) = \dim(\ker N^4) = 11$. Does the answer change if $\mathbb{C}$ is replaced by another field? How many classes are there if the number 11 is replaced by 13?

7. The minimal polynomial for an operator T on a vector space V is $(t - a)^3(t - b)^4$. Write the generalized eigenspace decomposition of V with respect to T . What are the possible values of dimension of V ? Is there a value of $\dim V$ for which the Jordan form matrix of T is uniquely decided by the given information? How many possible Jordan form matrices are there for T if $\dim V = 10$?

8. Problem 1b in 5.1.